EXP NO:20 Experiment to understand K-means clustering algorithm for a given data set.

AIM:
To understand the K-means clustering algorithm for the data set given.

PROGRAM:

python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

# Read the dataset
df = pd.read_csv("/content/student_admission.csv")
df.info()

# Display first 5 rows
df.head()

# Check missing values
print(df.isnull().sum())

# Fill missing feature values
df['Age'] = df['Age'].fillna(round(df['Age'].median()))
df['EntranceScore'] = df['EntranceScore'].fillna(round(df['EntranceScore'].mean()))

# Confirm no missing values remain
print(df[['Age', 'EntranceScore']].isnull().sum())

# Pairplot
sns.pairplot(df)

# Select features for clustering
features = df.iloc[:, [1, 2]].values

# Apply K-Means clustering
from sklearn.cluster import KMeans

model = KMeans(n_clusters=3, random_state=42, n_init=10)
model.fit(features)

# Create final dataset with cluster labels
Final = df.iloc[:, [1, 2]].copy()
Final["label"] = model.predict(features)

Final.head()

# Visualize clusters
sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=6) \
   .map(plt.scatter, "Age", "EntranceScore") \
   .add_legend()

plt.show()

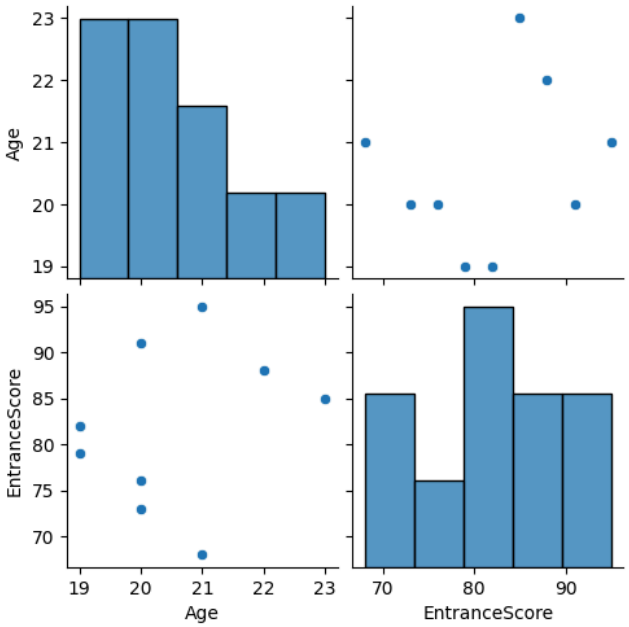
# Elbow method
features_el = df.iloc[:, [1, 2]].values

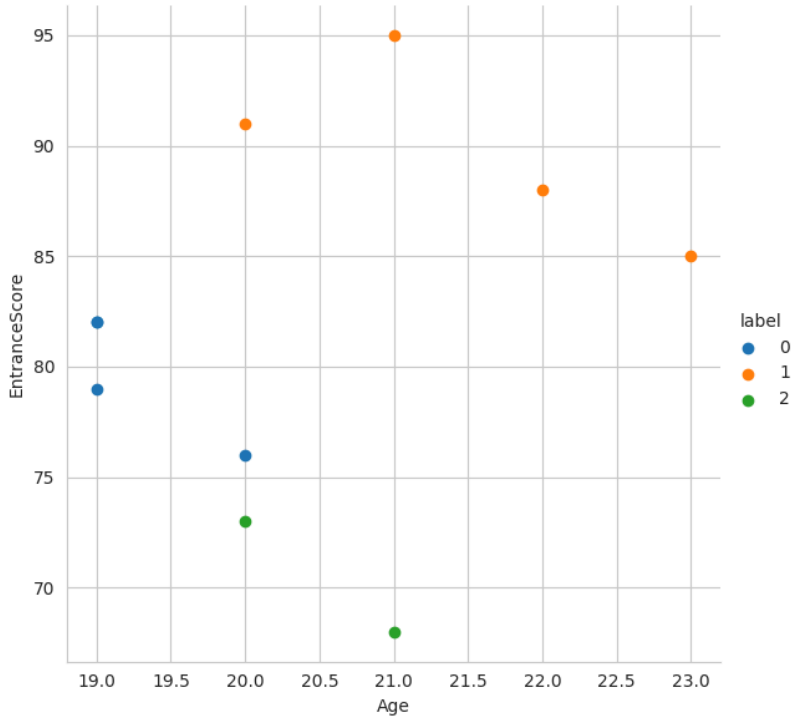
wcss = []

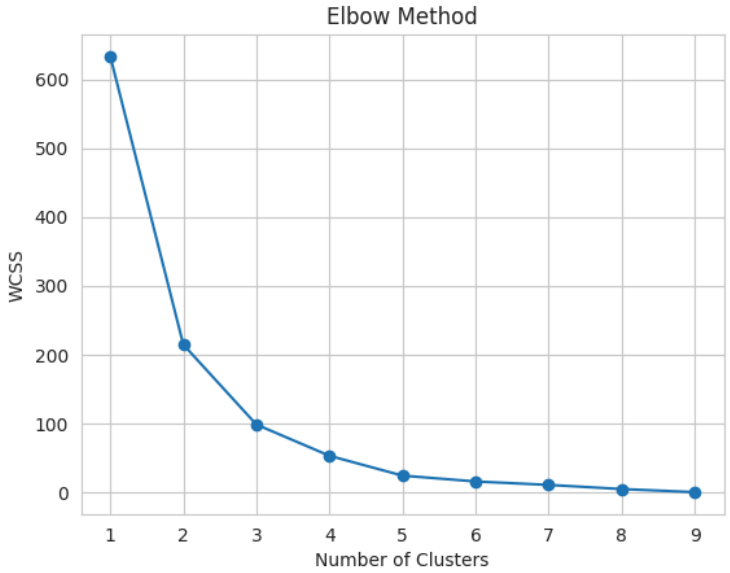
for i in range(1, 10):
   model = KMeans(n_clusters=i, random_state=42, n_init=10)
   model.fit(features_el)
   wcss.append(model.inertia_)

plt.plot(range(1, 10), wcss, marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

OUTPUT:


Age
EntranceScore
Age
EntranceScore


EntranceScore
Age
label
0
1
2



RESULT:

Thus the program to understand the K-means clustering algorithm for the given data set was executed successfully.